

AI IN LOGISTICS & SUPPLY CHAIN

AI-Driven Smart Retail Supply Chain Intelligence System

Let's start



OUR GOAL

analyze sales and supply chain patterns, forecast future demand, predict delivery delay risk, optimize inventory, and evaluate supplier and transportation performance



Dataset overview

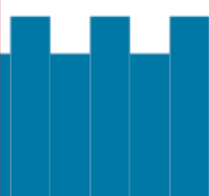
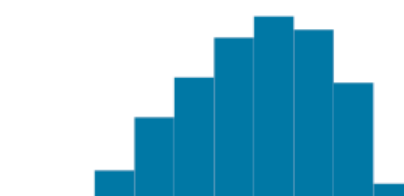
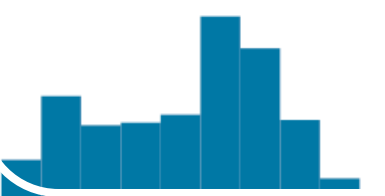
429K+ rows

For this project, we used the **Retail Sales Dataset** from Kaggle by Manjeet Singh.

The dataset includes three files: sales data, features data, and stores data.

The sales file contains weekly sales by store, department, date, and holiday status.
The features file includes external factors such as temperature, fuel price, CPI, unemployment, and markdown promotions.
The stores file includes store type and store size.

After merging all three files, we received one complete dataset that allowed us to analyze sales, demand patterns, store performance, promotions, holidays, and supply chain-related indicators.

	Date	Temperature	Fuel_Price	Markdown1
	182 unique values			NA51% 4855.310% Other (4030)49%
	05/02/2010	42.31	2.572	NA
	12/02/2010	38.51	2.548	NA
	19/02/2010	39.93		
	26/02/2010	46.63		
	05/03/2010	46.5		
	12/03/2010	57.79		
	19/03/2010	54.58		
	26/03/2010	51.45		
	02/04/2010	62.27		
	09/04/2010	65.86		
	16/04/2010	66.32		

```
# Column Non-Null Count Dtype
0 Store 421570 non-null int64
1 Dept 421570 non-null int64
2 Date 421570 non-null object
3 Weekly_Sales 421570 non-null float64
4 IsHoliday 421570 non-null bool
dtypes: bool(1), float64(1), int64(2), object(1)
memory usage: 13.3+ MB
```

```
Features dataset info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8190 entries, 0 to 8189
Data columns (total 12 columns):
# Column Non-Null Count Dtype
0 Store 8190 non-null int64
1 Date 8190 non-null object
2 Temperature 8190 non-null float64
3 Fuel_Price 8190 non-null float64
4 Markdown1 4032 non-null float64
5 Markdown2 2921 non-null float64
6 Markdown3 3613 non-null float64
7 Markdown4 3464 non-null float64
8 Markdown5 4050 non-null float64
9 CPI 7605 non-null float64
10 Unemployment 7605 non-null float64
11 IsHoliday 8190 non-null bool
dtypes: bool(1), float64(9), int64(1), object(1)
memory usage: 712.0+ KB
```

```
Stores dataset info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 45 entries, 0 to 44
Data columns (total 3 columns):
# Column Non-Null Count Dtype
0 Store 45 non-null int64
```

	Store	Dept	Date	Weekly_Sales	IsHoliday	Temperature	Fuel_Price	MarkDown1	MarkDown2	MarkDown3	MarkDown4	MarkDown5	CPI	Unemployment	Type	Size	Year	Month	Week	Quarter	Total_MarkDown	Sales_per_Size	IsHoliday_Int
0	1	1	2010-02-05	24924.50	False	42.31	2.572	0.0	0.0	0.0	0.0	0.0	211.096358	8.106	A	151315	2010	2	5	1	0.0	0.164719	0
1	1	1	2010-02-12	46039.49	True	38.51	2.548	0.0	0.0	0.0	0.0	0.0	211.242170	8.106	A	151315	2010	2	6	1	0.0	0.304263	1
2	1	1	2010-02-19	41595.55	False	39.93	2.514	0.0	0.0	0.0	0.0	0.0	211.289143	8.106	A	151315	2010	2	7	1	0.0	0.274894	0
3	1	1	2010-02-26	19403.54	False	46.63	2.561	0.0	0.0	0.0	0.0	0.0	211.319643	8.106	A	151315	2010	2	8	1	0.0	0.128233	0
4	1	1	2010-03-05	21827.90	False	46.50	2.625	0.0	0.0	0.0	0.0	0.0	211.350143	8.106	A	151315	2010	3	9	1	0.0	0.144255	0

Business problem

The main business problem is that retail companies need to manage demand, inventory, and logistics more accurately. If demand is underestimated, the company can face stockouts and lost sales. If demand is overestimated, it can lead to overstock and higher inventory costs.

At the same time, transportation costs, fuel prices, delivery delays, and supplier reliability can also affect supply chain performance. Because of this, the company needs a data-driven system that can forecast demand, identify risks, and support better operational decisions.

Our project solves this problem by combining Python-based analytics with Power BI dashboards.

Data cleaning and EDA

In the first stage, we cleaned and prepared the data.
We converted the Date column into datetime format, checked missing values and duplicates, and merged the three datasets into one final dataset.

Missing values in features dataset:

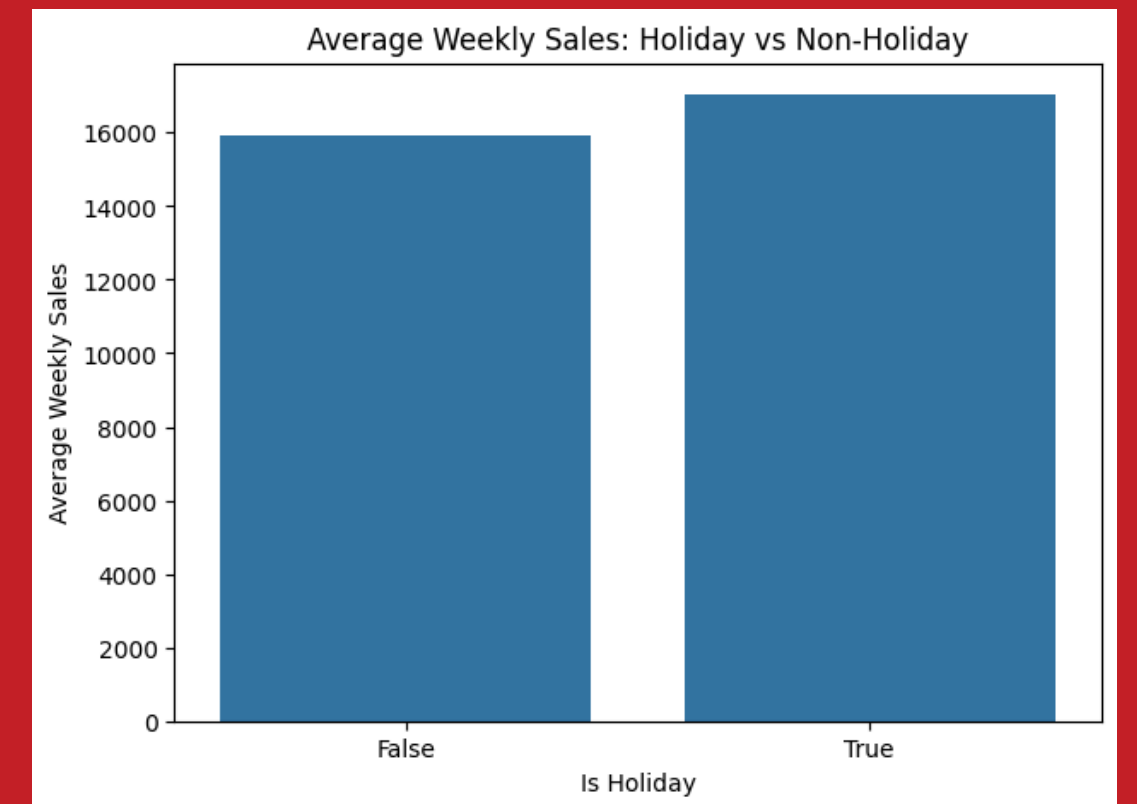
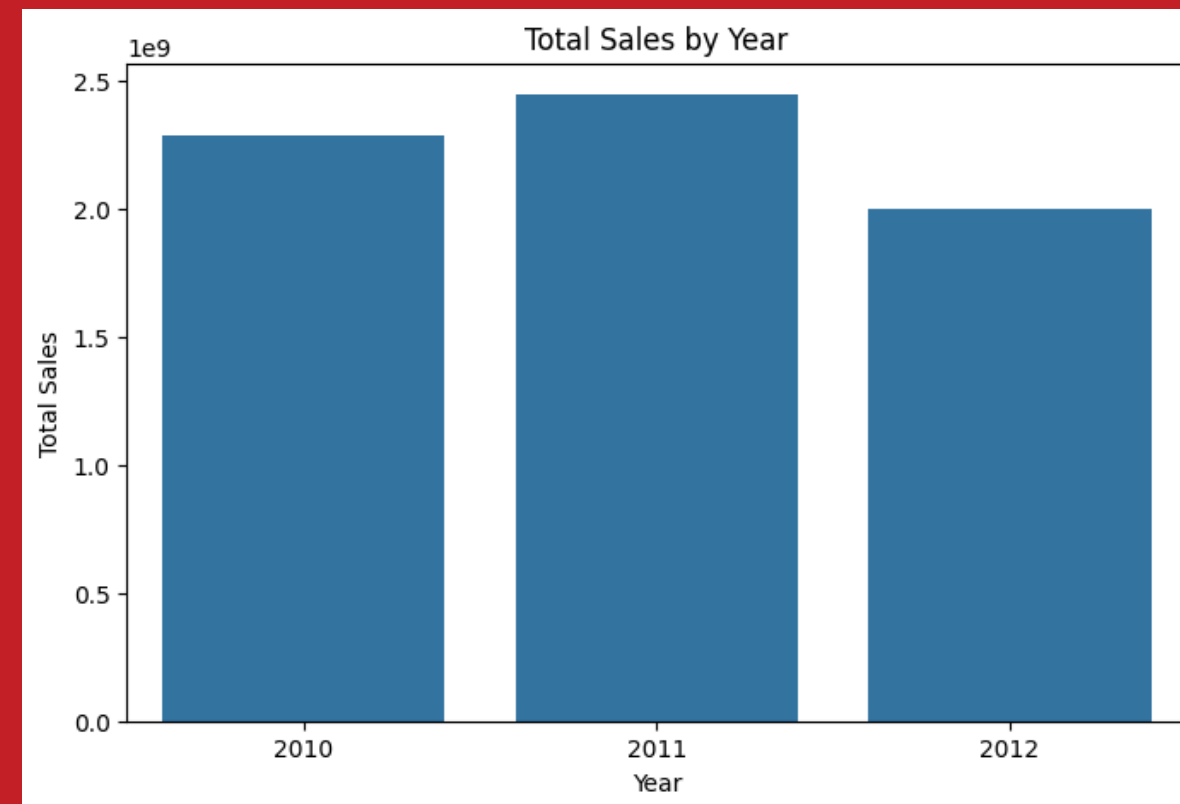
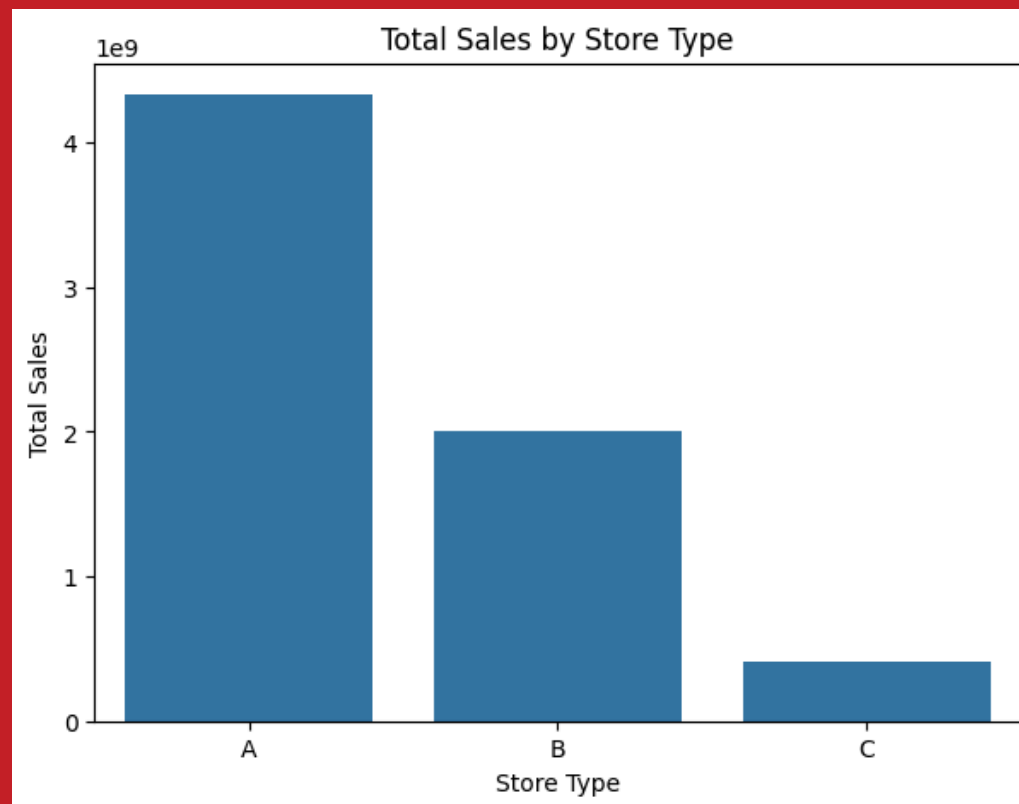
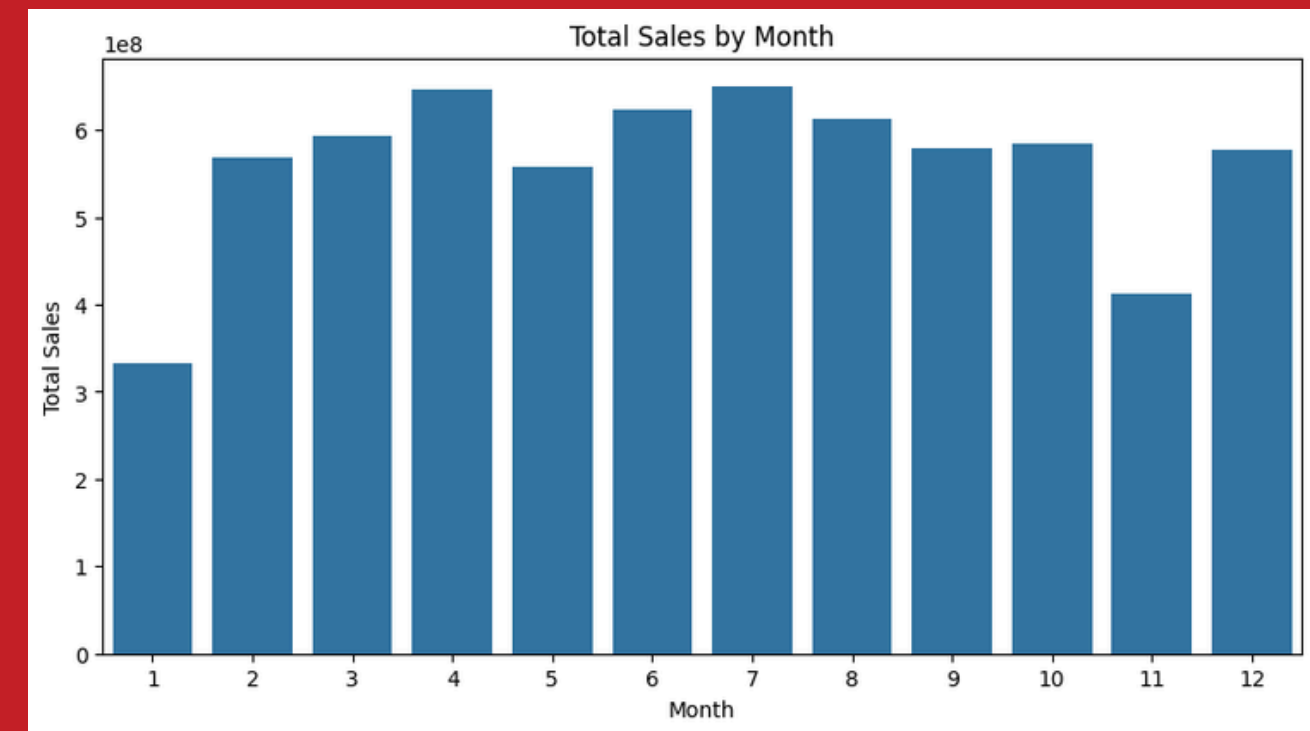
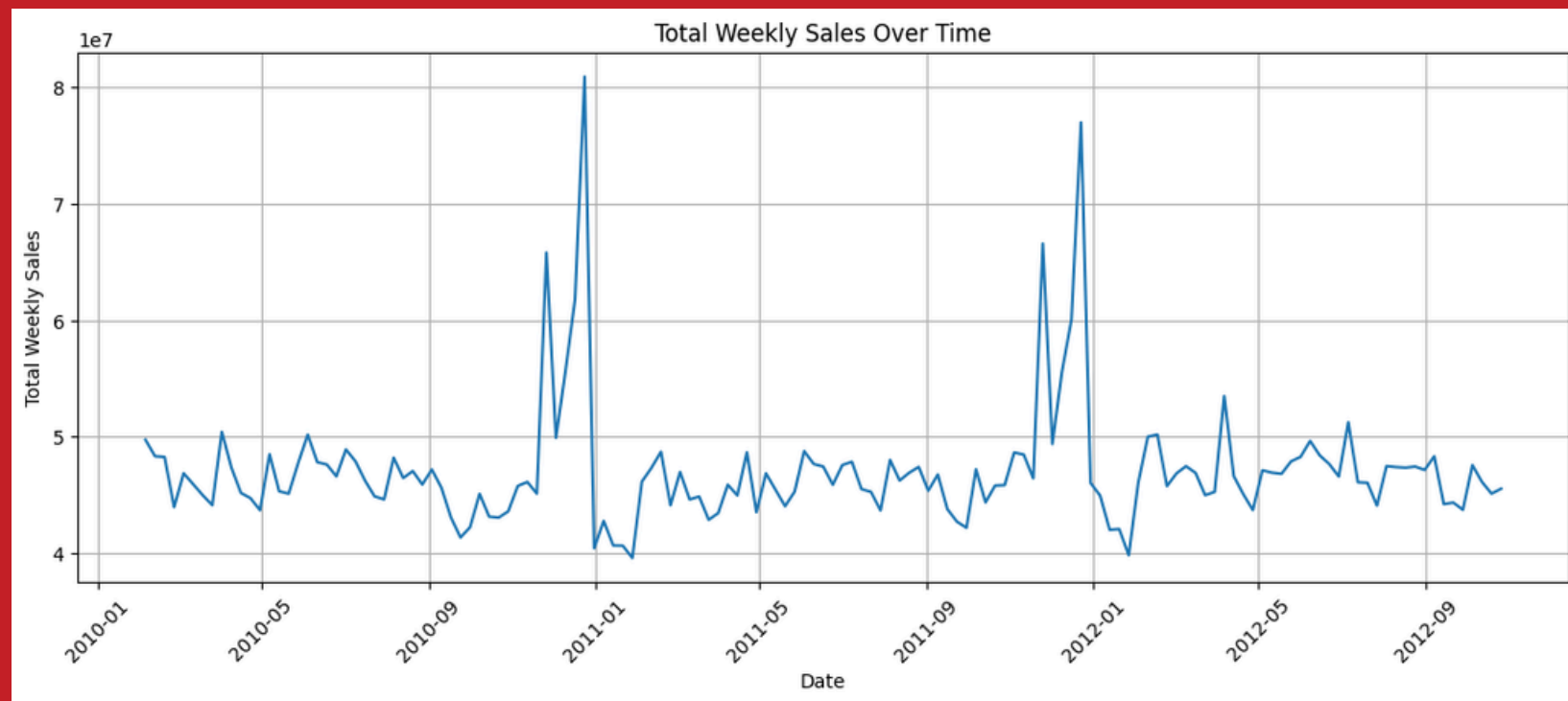
	0
Store	0
Date	0
Temperature	0
Fuel_Price	0
Markdown1	4158
Markdown2	5269
Markdown3	4577
Markdown4	4726
Markdown5	4140
CPI	585
Unemployment	585
IsHoliday	0

Missing values after cleaning:

	0
Store	0
Dept	0
Date	0
Weekly_Sales	0
IsHoliday	0
Temperature	0
Fuel_Price	0
Markdown1	0
Markdown2	0
Markdown3	0
Markdown4	0
Markdown5	0
CPI	0
Unemployment	0
Type	0
Size	0

Year	Month	Week	Quarter	Total_MarkDown	Sales_per_Size	IsHoliday_Int
2010	2	5	1	0.0	0.164719	0
2010	2	6	1	0.0	0.304263	1
2010	2	7	1	0.0	0.274894	0
2010	2	8	1	0.0	0.128233	0
2010	3	9	1	0.0	0.144255	0

Most missing values were in the markdown columns. We replaced them with 0 because missing markdown usually means there was no promotion. We also created new features such as Year, Month, Week, Quarter, Total Markdown, Sales per Size, and Holiday indicator.

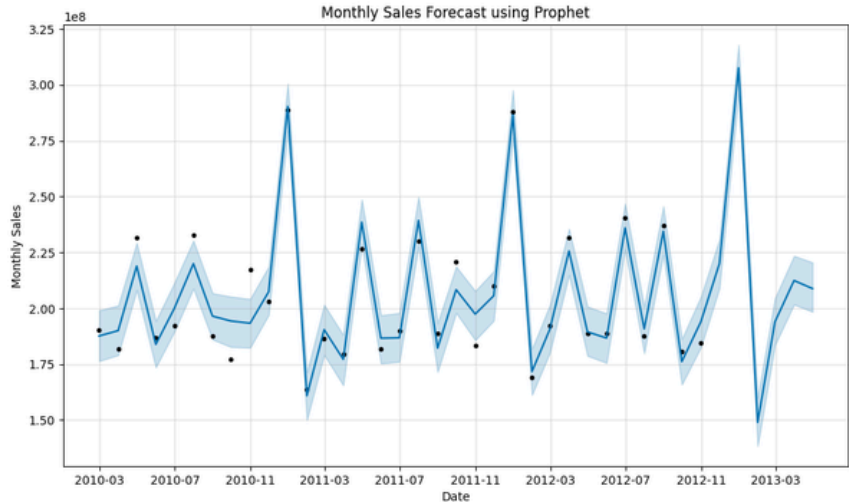
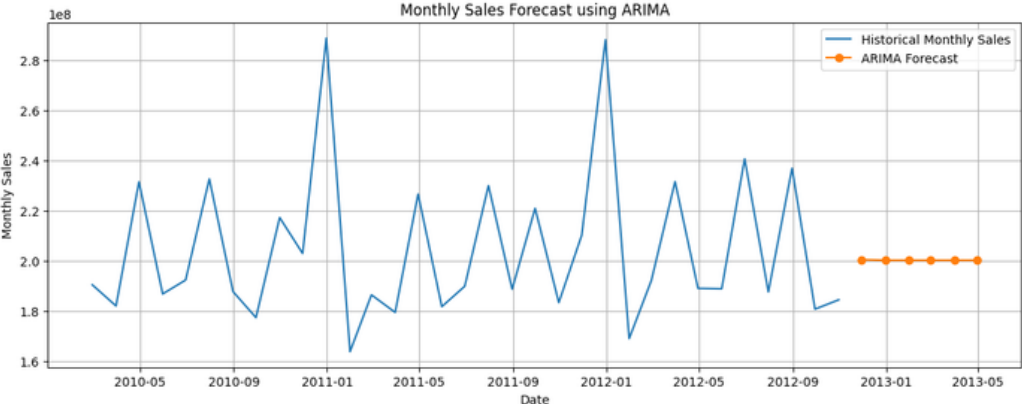


In the EDA part, we analyzed weekly sales trends, yearly and monthly sales, holiday impact, store types, top stores, top departments, correlations, and outliers.

The main finding was that sales are seasonal, holiday weeks have higher average sales, and Type A stores generate the highest total sales.

Demand forecasting models

Linear Regression, Random Forest Regressor, ARIMA, and Prophet



	Date	Monthly_Sales
0	2010-02-28	1.903330e+08
1	2010-03-31	1.819198e+08
2	2010-04-30	2.314124e+08
3	2010-05-31	1.867109e+08
4	2010-06-30	1.922462e+08
	Date	Monthly_Sales
28	2012-06-30	2.406103e+08
29	2012-07-31	1.875095e+08
30	2012-08-31	2.368508e+08
31	2012-09-30	1.806455e+08
32	2012-10-31	1.843617e+08

ARIMA and Prophet

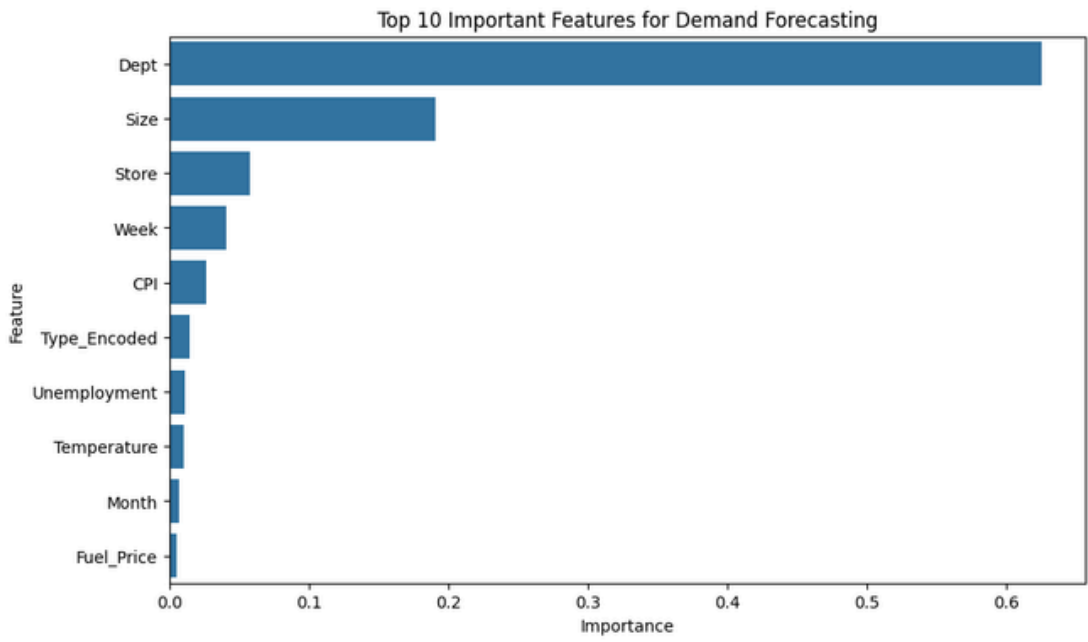
were used for monthly time-series forecasting. Prophet was also useful because it provided lower and upper forecast ranges. Based on the forecast, we created future inventory need levels.

Random Forest

performed much better and achieved an R2 score of about 0.973. This means it was able to explain most of the variation in weekly sales. The feature importance results showed that department, store size, store number, week, and CPI were important for predicting sales.

Linear Regression

was used as a simple baseline model, but its performance was weak. This shows that retail demand is not explained well by a simple linear relationship.

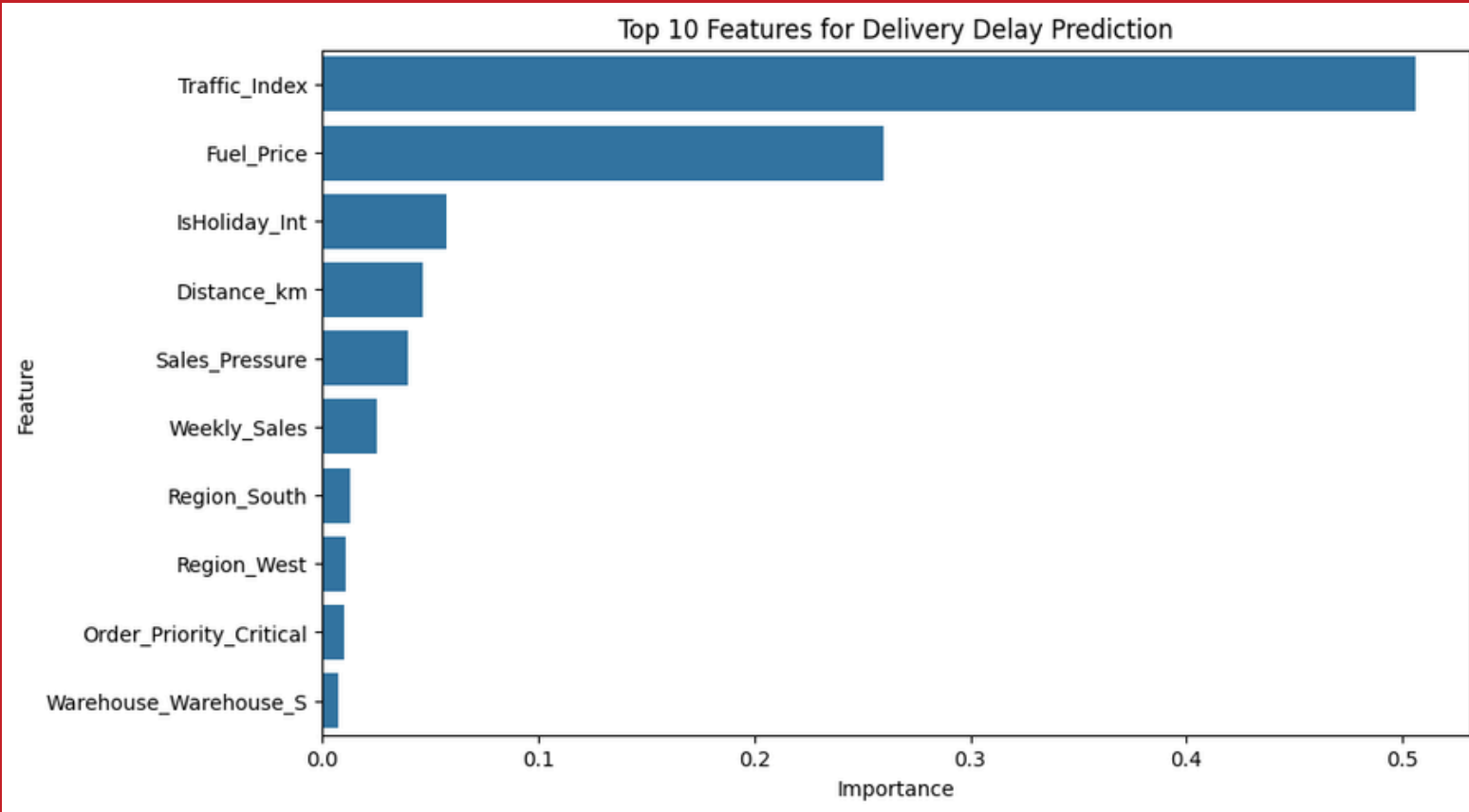


	Model	MAE	RMSE	R2 Score
0	Linear Regression	14551.750602	21790.260986	0.089469
1	Random Forest Regressor	1456.408365	3721.652198	0.973439

Store numbers were grouped into regions. Each region was connected to a warehouse. Store type was used as a proxy for shipping mode. Weekly sales level was used to define order priority. Traffic index was estimated using fuel price, holiday weeks, and sales pressure.

	Store	Dept	Date	Weekly_Sales	Region	Warehouse	Shipping_Mode	Distance_km	Order_Priority	Traffic_Index
0	1	1	2010-02-05	24924.50	East	Warehouse_E	Standard Class	78	Critical	0.037180
1	1	1	2010-02-12	46039.49	East	Warehouse_E	Standard Class	78	Critical	0.244469
2	1	1	2010-02-19	41595.55	East	Warehouse_E	Standard Class	78	Critical	0.035109
3	1	1	2010-02-26	19403.54	East	Warehouse_E	Standard Class	78	High	0.031812
4	1	1	2010-03-05	21827.90	East	Warehouse_E	Standard Class	78	Critical	0.046027

	Model	Accuracy	Precision	Recall	F1 Score
0	Logistic Regression	0.941647	0.872621	0.848072	0.860172
1	Decision Tree	0.999976	0.999944	0.999944	0.999944
2	Random Forest Classifier	0.999976	0.999888	1.000000	0.999944



Delivery delay prediction

Then we created the Delivery Delay target using the logistic probability formula. We trained three models: Logistic Regression, Decision Tree, and Random Forest Classifier. The results showed that Traffic Index, fuel price, holiday weeks, distance, and sales pressure were the main delay drivers. This helps identify high-risk delivery routes.

we created a proxy logistics layer based on the available data

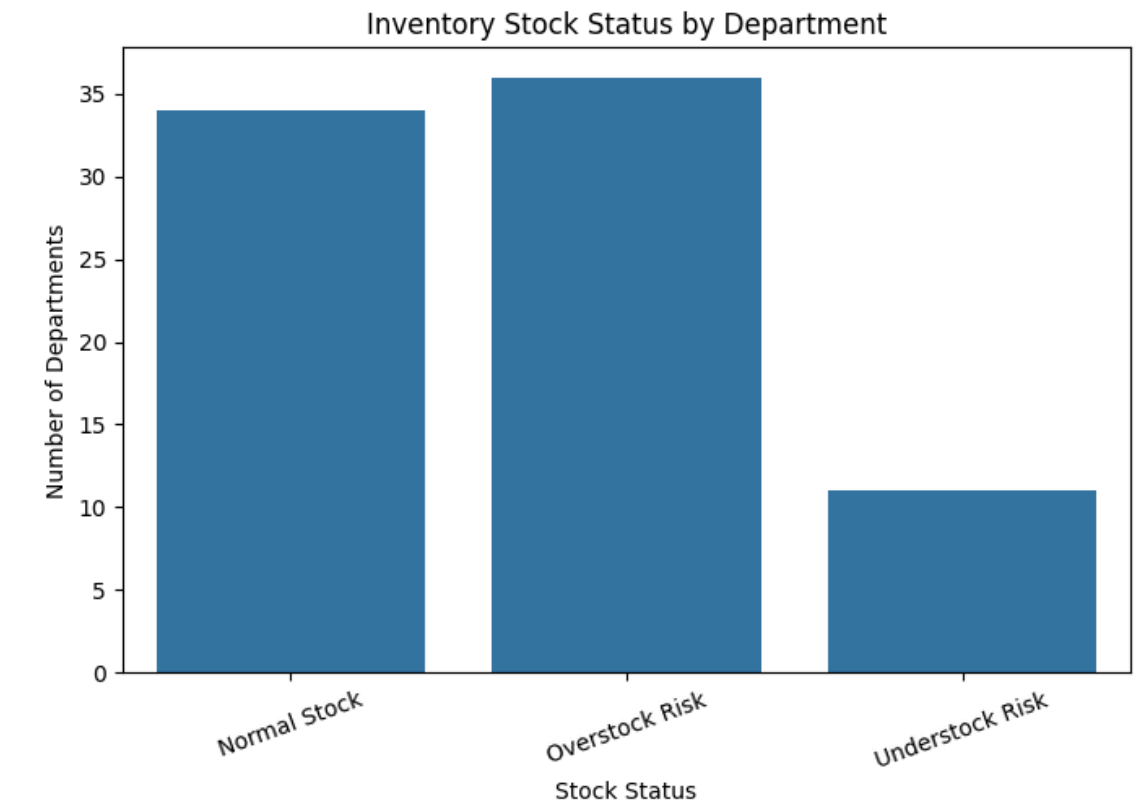
Inventory optimization

ABC Analysis and K-Means Clustering

Since the dataset did not include direct product stock levels, we used departments as product categories and weekly sales as a demand proxy

	ABC_Category	Number_of_Departments	Total_Sales	Average_Weekly_Sales	Sales_Share_%
0	A	28	5.344872e+09	30110.736544	79.333509
1	B	20	1.046253e+09	9083.576458	15.529448
2	C	33	3.460938e+08	3387.746438	5.137043

	Cluster	Number_of_Departments	Total_Sales	Average_Weekly_Sales	Average_Demand_Volatility	Average_Store_Coverage	Average_MarkDown
0	0	9	2.141413e+07	5922.850711	-3.626798	20.111111	12964.585619
1	1	61	3.424786e+09	9260.443142	0.878393	43.524590	6642.545290
2	2	11	3.291019e+09	47125.008800	0.630472	45.000000	6336.781958

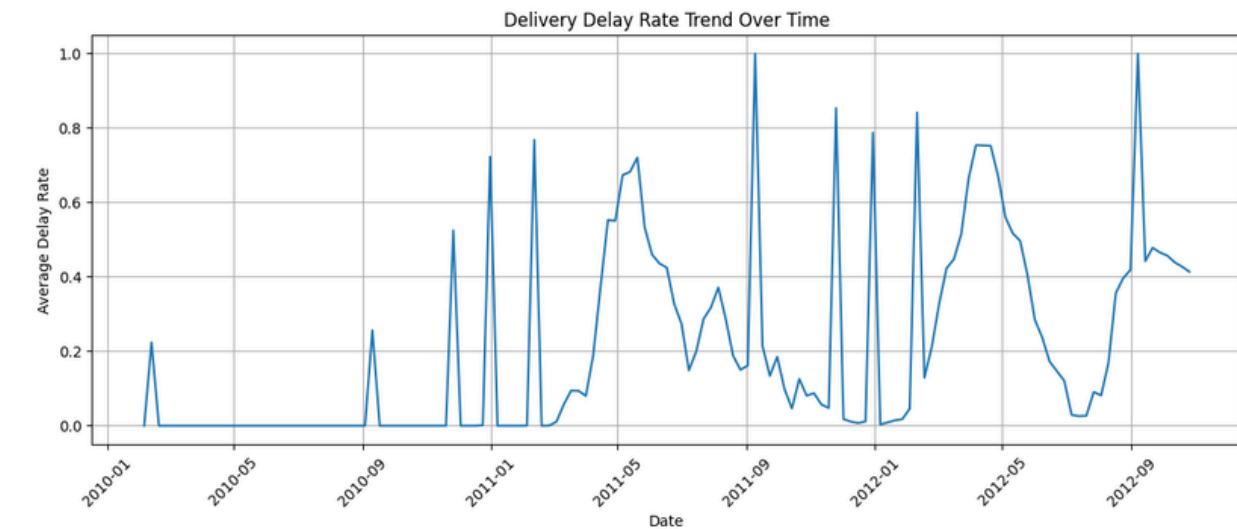
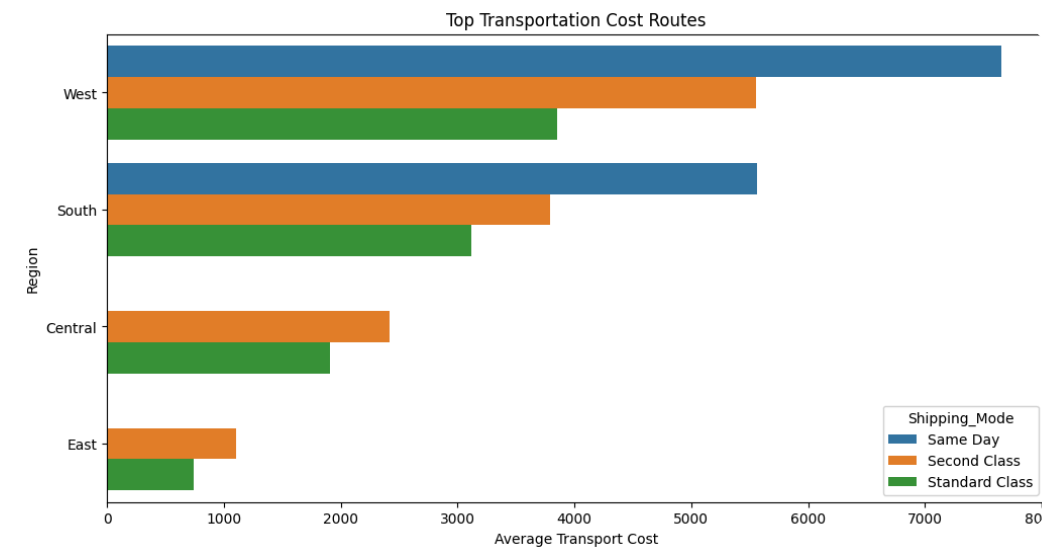
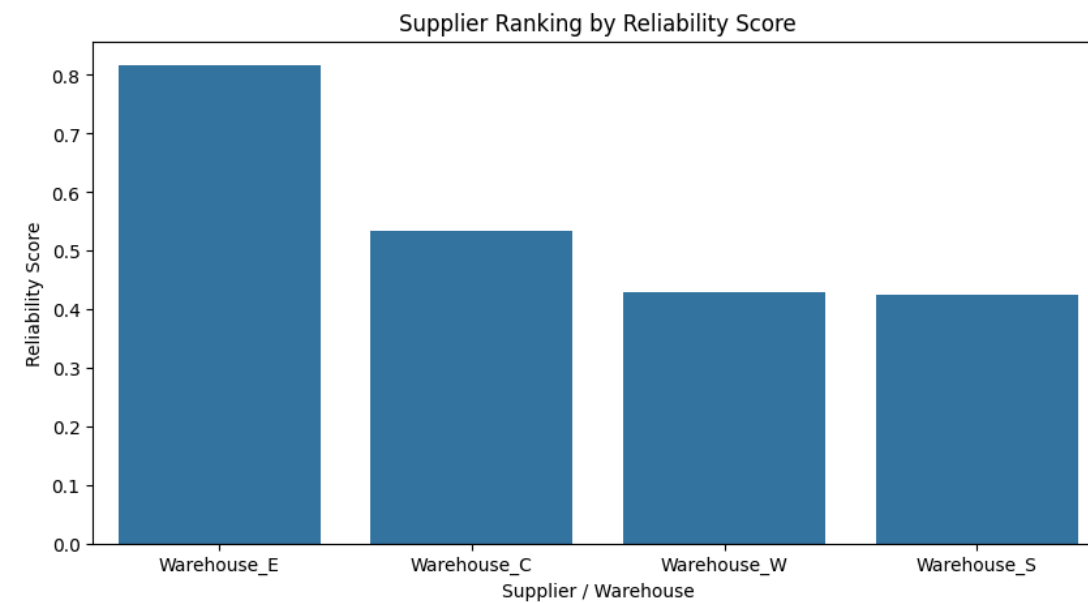


ABC Analysis showed that Category A departments generated around 79.33% of total sales. This means a small group of departments is responsible for most of the revenue. These departments should be monitored more carefully.

K-Means Clustering grouped departments based on total sales, average weekly sales, demand volatility, store coverage, and average markdown. We also identified overstock and understock risks and calculated reorder points using average weekly demand, lead time, and safety stock.

Supplier & transportation analytics

again used a proxy supply chain network because the dataset did not contain real supplier or route data

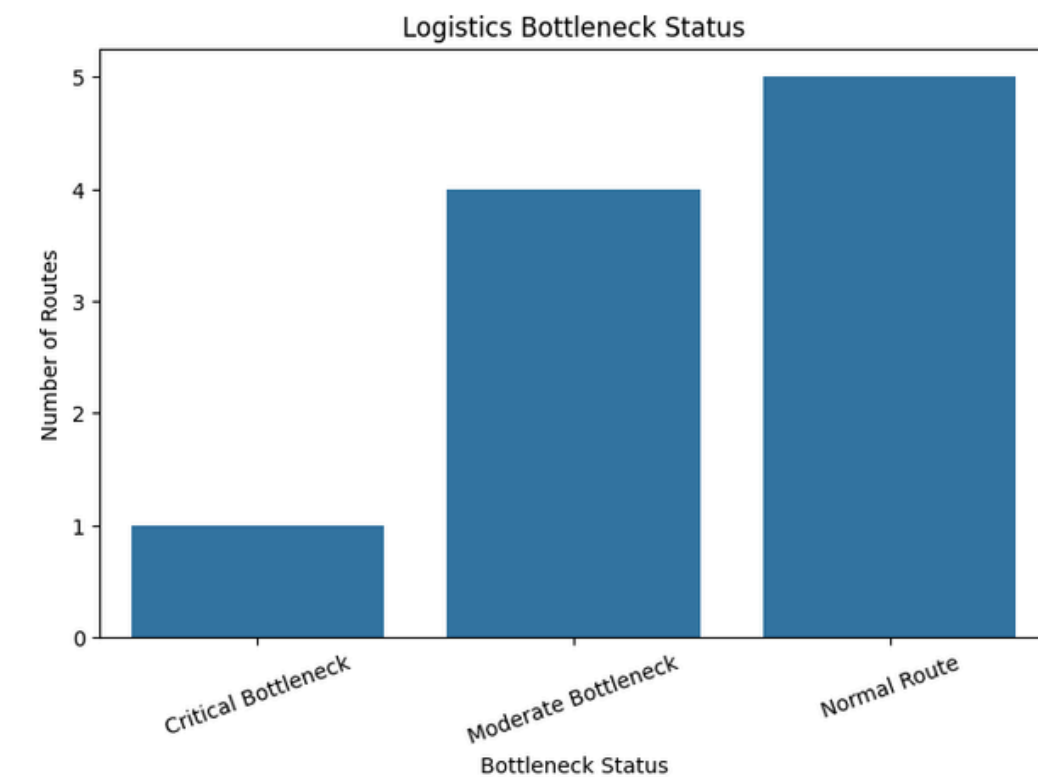


Warehouses were treated as supplier or distribution partners. Regions represented transport zones, and shipping modes represented delivery service types.

We estimated transportation cost using distance, fuel price, shipping mode, and traffic index.

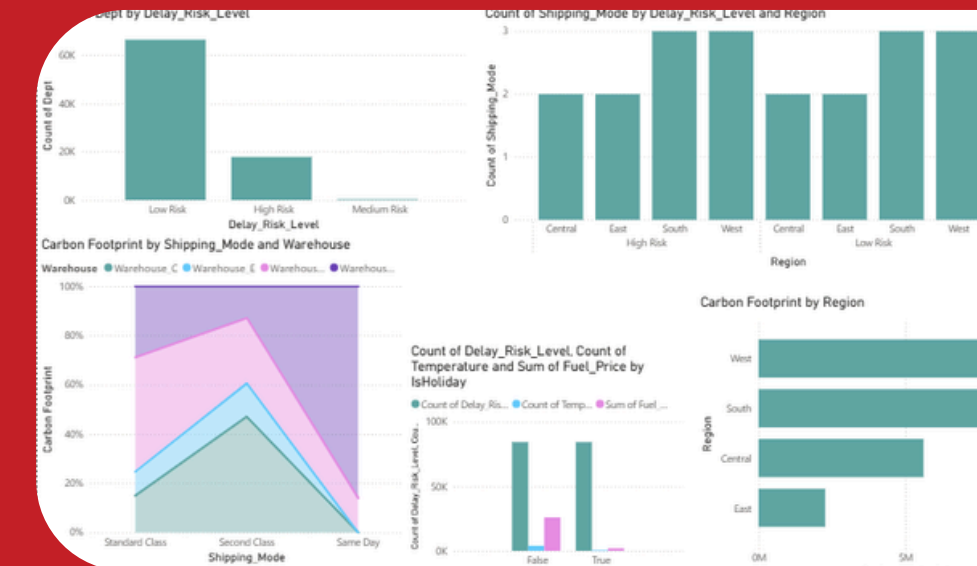
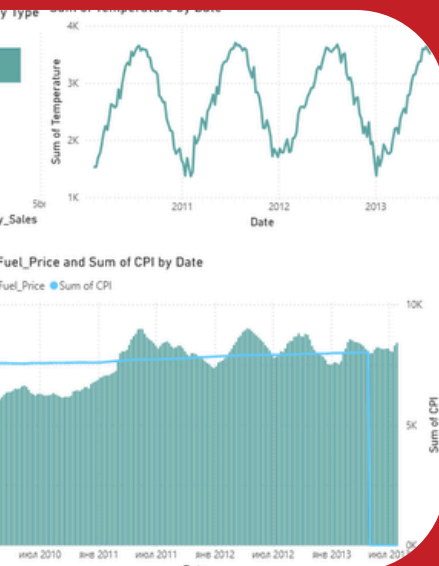
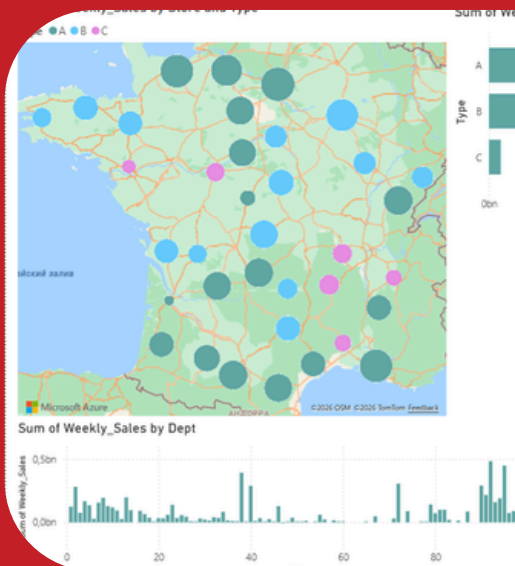
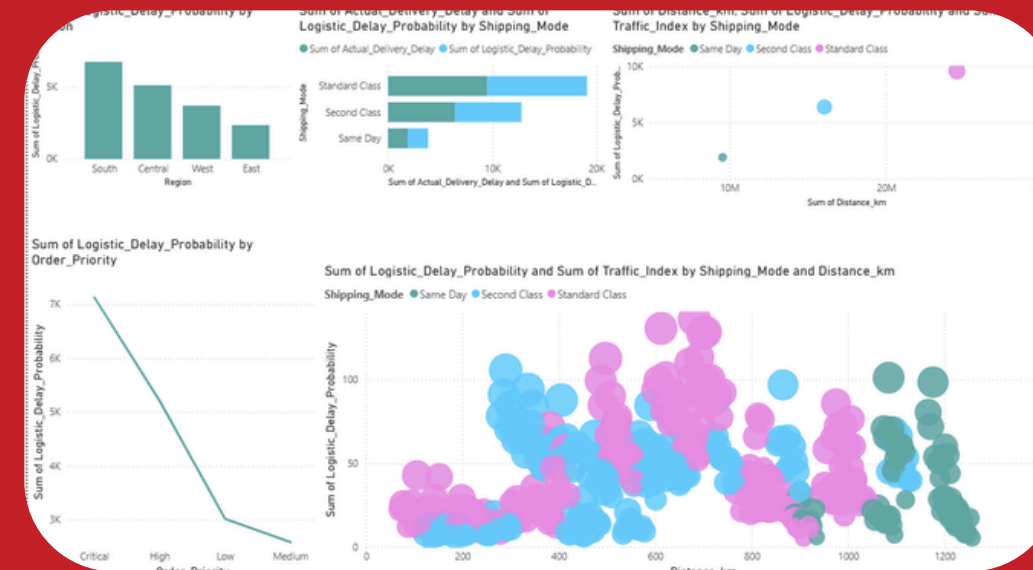
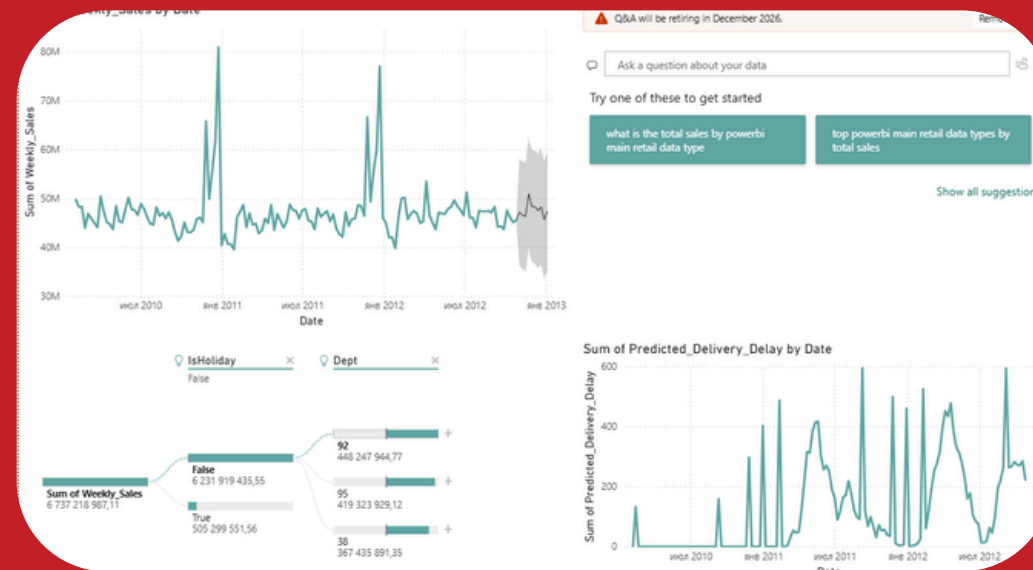
The supplier ranking showed that Warehouse_E had the highest reliability score, while Warehouse_S had the lowest reliability.

Transportation analysis showed that West and South routes were more expensive and riskier, especially with Same Day and Second Class shipping. We also identified logistics bottlenecks and high-risk supply chain routes.



Power BI

Vizualization



Key business insights

1. Sales are seasonal and increase during holiday periods. This means the company should prepare more inventory and logistics capacity before high-demand weeks.
2. Random Forest was the strongest model for weekly demand forecasting. This shows that demand patterns are complex and non-linear.
3. Category A departments generate most of the revenue, so they should receive priority in inventory planning.
4. Delivery delay risk increases when traffic, fuel price, distance, holiday pressure, and order priority are high.
5. West and South routes have higher transportation cost and supply chain risk.

Recommendations

- Based on the results, the company should use machine learning models, especially Random Forest, for weekly demand forecasting.
- The company should monitor Category A departments more closely and maintain enough safety stock for high-demand and high-volatility departments.
- High-risk delivery routes should be monitored before holidays and peak sales periods.
- The company can prepare earlier shipments and increase transport capacity for critical orders.
- Warehouse_S and Warehouse_W routes should be reviewed because they show higher risk or lower reliability.
- Fuel price should also be included in logistics budgeting because it strongly affects transportation cost.

Conclusion

To conclude, this project developed an AI-driven retail supply chain intelligence system using a retail sales dataset from Kaggle.

- The project included data cleaning, exploratory data analysis, demand forecasting, delivery delay prediction, inventory optimization, and supplier and transportation analytics.
- The results show that machine learning and business intelligence can support better supply chain decisions.
- Even though some logistics variables were not directly available, proxy features allowed us to build a complete analytical pipeline and prepare the results for Power BI reporting.

Thank you!

the kaggle dataset